Vectors-1

→ Multiplication of a scalar by a vector yield another vector of diff. mag. or opp. dirⁿ. Thus,

Collinear vector : $\boxed{a = \lambda b}$ $(\lambda \neq 0)$

→ To get ~~two~~ vectors b/w two points, do

$$\boxed{\text{Final} - \text{Initial}}$$

→ Vectors can be written in terms of orthogonal triad

e.g. $\boxed{\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}}$

Triangle Law of Addition :

Arrangement ⇒ Tail to head joining

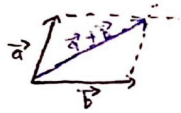
Resultant ⇒ Vector from open tail to open head.



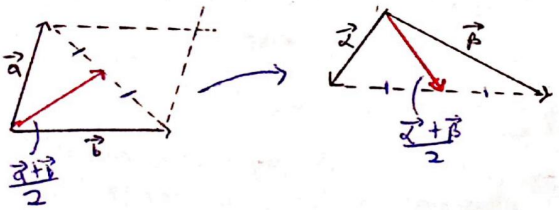
Parallelogram law of Addition :

Arrangement ⇒ Tail to Tail joining

Resultant ⇒ Along Diagonal of ||gm



Special Result ⇒ Mid-point of diagonal is $\frac{\vec{a}+\vec{b}}{2}$ (vector).

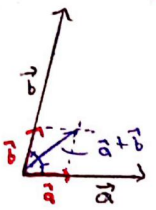


(*) In a ||gm, diagonals bisect each other.

Vector along angle bisector ⇒

1) Calculate unit vectors of $\vec{a}$ & $\vec{b}$

2) Vector addⁿ of $\hat{a}$ & $\hat{b}$ gives required vector



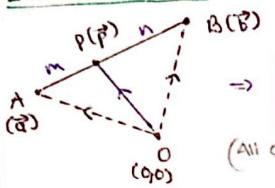
$\boxed{\vec{a}+\vec{b} = \vec{v}_1}$

It is not mandatory that sum of two unit vectors must be a unit vector.

→ Thus, $\vec{v}_1$ may or may not be an unit vector.

Section Formula :

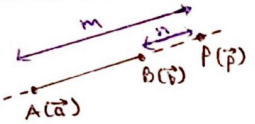
Internal Division ⇒



$\Rightarrow \boxed{\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}}$

(All are position vectors)

External Division ⇒

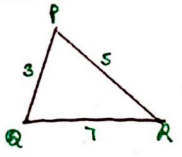


$\boxed{\vec{r} = \frac{m\vec{b} - n\vec{a}}{m-n}}$

Cosine Formula :

→ Used when all 3 sides of △ are known & 1 angle are unknown.

$$\cos\theta = \frac{(\text{Sum of sq. of sides forming } \theta) - (\text{Sq. of third side})}{2\,(\text{Product of sides forming } \theta)}$$



$\cos P = \frac{3^2+5^2-7^2}{2(3)(5)}$

$\cos R = \frac{5^2+7^2-3^2}{2(5)(7)}$

$\cos Q = \frac{3^2+7^2-5^2}{2(3)(7)}$

Dot Product :

$$\boxed{\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta}$$

$\theta \Rightarrow$ Angle b/w $\vec{a}$ & $\vec{b}$

Important Properties ⇒

1) $\vec{a}\cdot\vec{b}$ = +ve ⇒ θ is acute

$\vec{a}\cdot\vec{b} = 0 \Rightarrow \theta$ is 90° $(\vec{a} \perp \vec{b})$

$\vec{a}\cdot\vec{b}$ = -ve ⇒ θ is obtuse

2) $\vec{a}\cdot\vec{a} = |\vec{a}|^2 = \vec{a}^2$

3) $(\vec{a} \pm \vec{b})^2 = |\vec{a}|^2 + |\vec{b}|^2 \pm 2\vec{a}\cdot\vec{b}$

$(\vec{a}-\vec{b})\cdot(\vec{a}+\vec{b}) = |\vec{a}|^2 - |\vec{b}|^2$

$(\vec{a}+\vec{b}+\vec{c})^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$

Angle b/w two Vectors :

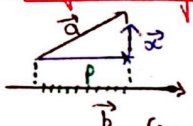
→ Make initial points of two vectors same to measure angle.



→ $\vec{a} \wedge \vec{b} = \cos\theta = \boxed{\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}}$

→ $\theta \in [0, \pi]$ (Always lie in 1st or 2nd quadrant)

Projection of a Vector :



Projection of $\vec{a}$ on $\vec{b}$ = p

(Only length) $\boxed{p = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}}$

(*) Projection vector of $\vec{a}$ along $\vec{b}$ / Vector component of $\vec{a}$ along $\vec{b}$

$$\boxed{\vec{p} = (|\vec{a}|\cos\theta)\,\hat{b}}$$

(*) Ion component of $\vec{a}$ w.r.t ~~of~~ $\vec{b}$ is

$$\boxed{\vec{x} = \vec{a} - (|\vec{a}|\cos\theta)\,\hat{b}}$$

Important Conceptual Point ⇒

1) $\boxed{\vec{a}\cdot\vec{b} = \vec{a}\cdot\vec{c}} \not\Rightarrow \vec{b} = \vec{c}$

$\vec{a}\cdot\vec{b} - \vec{a}\cdot\vec{c} = 0$

$\boxed{\vec{a}\cdot(\vec{b}-\vec{c}) = 0 \Rightarrow \vec{a} \perp (\vec{b}-\vec{c})}$

3) Vector Relation is gold-mine of info

Dot with any vector ← $\boxed{\vec{x} = \vec{a} + 2\vec{b}}$ → squaring

$\vec{x}\cdot\vec{a} = \vec{a}\cdot\vec{a} + 2\vec{a}\cdot\vec{b}$

$|\vec{x}|^2 = |\vec{a}|^2 + 4|\vec{b}|^2 + 4\vec{a}\cdot\vec{b}$

Cross with any vector

$\vec{x}\times\vec{c} = \vec{a}\times\vec{c} + 2\vec{b}\times\vec{c}$

Sq. lene se go 2 vector same side hote hai, unlike beech dot aaya hai

Cross Product :

$$\vec{a} \times \vec{b} = |\vec{a}|\,|\vec{b}| \sin\theta\, \hat{n}$$

Vector $(\vec{a} \times \vec{b})$ is ⊥ar to both $\vec{a}$ & $\vec{b}$

Important Point ⇒

1) $(\vec{a} \times \vec{b}) = -(\vec{b} \times \vec{a})$

2) $m\vec{a} \times n\vec{b} = mn(\vec{a} \times \vec{b})$ $(m, n \in \text{Const.})$

3) $\vec{a} \times (\vec{b} \pm \vec{c}) = (\vec{a} \times \vec{b}) \pm (\vec{a} \times \vec{c})$
$(\vec{a} \pm \vec{b}) \times \vec{c} = (\vec{a} \times \vec{c}) \pm (\vec{b} \times \vec{c})$
Position matters

4) If $\vec{a} \times \vec{b} = 0 \Rightarrow \vec{a} \parallel \vec{b}$
$\vec{a} \times \vec{b} = |\vec{a}| \cdot |\vec{b}|\, \hat{n} \Rightarrow \vec{a} \perp \vec{b}$
$(\sin\theta = 1)$

5) 

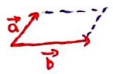
$i \times j = k$ $\quad j \times k = i$ $\quad k \times i = j$
$j \times i = -k$ $\quad k \times j = -i$ $\quad i \times k = -j$

6) Vector ⊥ar to $\vec{a}$ & $\vec{b}$ $= \lambda(\vec{a} \times \vec{b})$
Unit vector ⊥ar to $\vec{a}$ & $\vec{b}$ $= \dfrac{\pm(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|}$

7) If $(\vec{a} \times \vec{b}) = (\vec{a} \times \vec{c})$
$(\vec{a} \times \vec{b}) - (\vec{a} \times \vec{c}) = 0 \Rightarrow \vec{a} \times (\vec{b} - \vec{c}) = 0$
$\Rightarrow \vec{a} \parallel (\vec{b} - \vec{c})$ $\quad \vec{a} = \lambda(\vec{b} - \vec{c})$ or $(\vec{b} - \vec{c}) = \mu\vec{a}$

Different Areas ⇒

Parallelogram :

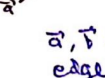


Area $= |\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$

Triangle :



area $= \dfrac{|\vec{a} \times \vec{b}|}{2}$



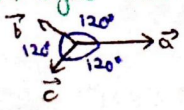
Area $= \frac{1}{2}|(\vec{c} - \vec{a}) \times (\vec{b} - \vec{a})|$
$= \frac{1}{2}|(\vec{a} \times \vec{b}) + (\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a})|$

Important Inequality :

If $\vec{a}, \vec{b}, \vec{c}$ = 3 unit vectors then

$$(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq -\tfrac{3}{2}$$

For equality, angles must be 120°



Scalar Triple Product :

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = [\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

(*) Dot & cross can change positions
$(\vec{a} \times \vec{b}) \cdot \vec{c} = \vec{a} \cdot (\vec{b} \times \vec{c})$

Important Properties :⇒

1) In cyclic order, STP remains unchanged.
$[a\ b\ c] = [b\ c\ a] = [c\ a\ b]$
$[a\ b\ c] = -[a\ c\ b]$

2) If any 2 or all 3 vectors are parallel, then
$[a\ b\ c] = 0$

3) $[\hat{i}\ \hat{j}\ \hat{k}] = 1$ & $[\hat{i}\ \hat{k}\ \hat{j}] = -1$

Major Formulas :⇒

1) $[\lambda a\ \beta b\ \gamma c] = \lambda\beta\gamma\,[a\ b\ c]$

Imp
2) $[\lambda a + \beta b + \gamma c\ \ la + mb + nc\ \ pa + qb + rc]$
$= \begin{vmatrix} \lambda & \beta & \gamma \\ l & m & n \\ p & q & r \end{vmatrix} [a\ b\ c]$

3) $[l\ m\ n][a\ b\ c] = \begin{vmatrix} l\cdot a & l\cdot b & l\cdot c \\ m\cdot a & m\cdot b & m\cdot c \\ n\cdot a & n\cdot b & n\cdot c \end{vmatrix}$

4) $[\vec{a} \times \vec{b}\ \ \vec{b} \times \vec{c}\ \ \vec{c} \times \vec{a}] = [a\ b\ c]^2$

Geometrical Meaning :⇒

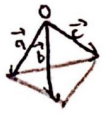
1) Parallelopiped ⇒



Volume = Base Area × Height
Volume $= [\vec{a}\ \vec{b}\ \vec{c}]$

$\vec{a}, \vec{b}$ & $\vec{c}$ must be coinitial edge vectors.

2) Tetrahedron ⇒



Volume $= \frac{1}{3} \times$ Base Area $\times$ Height
Volume $= \frac{1}{6}[\vec{a}\ \vec{b}\ \vec{c}]$

Vector Triple Product :

$(\vec{a} \times \vec{b}) \times \vec{c}$ is coplanar with $\vec{a}$ & $\vec{b}$ and ⊥ar to $\vec{c}$.

$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$$

Important Properties :⇒

1) V.T.P is a vector quantity

2) $\vec{a} \times (\vec{b} \times \vec{c}) \neq (\vec{a} \times \vec{b}) \times \vec{c}$

3) Unit vector coplanar with $\vec{a}$ & $\vec{b}$ & ⊥ar to $\vec{c}$ is $\pm \dfrac{(\vec{a} \times \vec{b}) \times \vec{c}}{|(\vec{a} \times \vec{b}) \times \vec{c}|}$

Product of 4 vectors :

Lagrange's Identity :

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = \begin{vmatrix} \vec{a} \cdot \vec{c} & \vec{a} \cdot \vec{d} \\ \vec{b} \cdot \vec{c} & \vec{b} \cdot \vec{d} \end{vmatrix}$$

Cross Product :

$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [a\ b\ d]\vec{c} - [a\ b\ c]\vec{d}$
$= [c\ d\ a]\vec{b} - [c\ a\ b]\vec{d}$

Theorem in Plane :

Condⁿ of Coplanarity ⇒ $[\vec{a}\ \vec{b}\ \vec{c}] = 0$

→ Two vectors are always coplanar

→ If $\vec{a}$ is coplanar with $\vec{b}$ & $\vec{c}$, then all three are linearly dependent on each other.

$\vec{a} = \lambda\vec{b} + \mu\vec{c}$; $\vec{b} = \lambda\vec{a} + \mu\vec{c}$; $\vec{c} = \lambda\vec{a} + \mu\vec{b}$

Theorem in Plane :

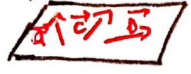
→ Theorem states that if $\vec{a}$ is coplanar with $\vec{b}$ & $\vec{c}$, then one can assume $\vec{a}$ in terms of $\vec{b}$ & $\vec{c}$ as:

$\vec{a} = \lambda\vec{b} + \mu\vec{c}$
$\vec{b} = \lambda\vec{a} + \mu\vec{c}$
$\vec{c} = \lambda\vec{a} + \mu\vec{b}$

→ Linearly Dependent

→ Three vectors are coplanar if

$[\vec{a}\ \vec{b}\ \vec{c}] = 0$



NOTE : Two vectors are always Coplanar.

Theorem in Space :

→ It states that any vector can be assumed in terms of three non-coplanar vectors as:

$$\vec{x} = \lambda\vec{a} + \mu\vec{b} + \delta\vec{c}$$

Means 4 or more than 4 vectors are always linearly dependent.

Vectors-2

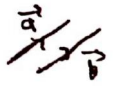
→ For any three non-coplanar vectors



$$[\vec{a}\ \vec{b}\ \vec{c}] \neq 0$$

ⓐ Linearly Dependent :→

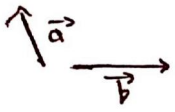
→ Simply, collinear or Coplanar vectors are linearly dependent



$$\vec{b} = \lambda \vec{a}$$

ⓑ Linearly Independent :→

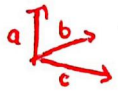
→ Non-collinear vectors are linearly independent.



Mean, we can write

$$x\vec{a} = y\vec{b}$$ only if $$x = y = 0$$

→ Thus, for three non-coplanar vectors

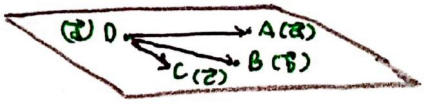


$$x\vec{a} = y\vec{b} + z\vec{c}$$

if & only if $x = y = z = 0$

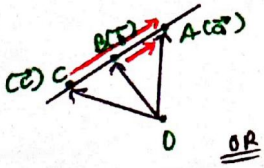
Important Note :

1] Three points are always Coplanar if 4 points with given position vectors are coplanar.



$$[\overrightarrow{DA}\ \overrightarrow{DB}\ \overrightarrow{DC}] = 0$$

2] Two points are always collinear if 3 points with given position vectors are collinear.



$\overrightarrow{CA} \parallel \overrightarrow{BA}$

$$\overrightarrow{CA} = \lambda \overrightarrow{BA}$$

OR

$$[\vec{a}\ \vec{b}\ \vec{c}] = 0$$

Reciprocal System of Vectors :

→ If $\vec{a}$, $\vec{b}$ & $\vec{c}$ are three non-collinear, non-coplanar vectors, then :

$$\vec{a}' = \frac{\vec{b} \times \vec{c}}{[\vec{a}\ \vec{b}\ \vec{c}]} \qquad \vec{b}' = \frac{\vec{c} \times \vec{a}}{[\vec{a}\ \vec{b}\ \vec{c}]}$$

$$\vec{c}' = \frac{\vec{a} \times \vec{b}}{[\vec{a}\ \vec{b}\ \vec{c}]}$$

Here, systems $(\vec{a}, \vec{b}, \vec{c})$ and $(\vec{a}', \vec{b}', \vec{c}')$ are reciprocal system of each other.

NOTE :

1] $$[\vec{a}\ \vec{b}\ \vec{c}] = \frac{1}{[\vec{a}'\ \vec{b}'\ \vec{c}']}$$

2] $$\vec{a}.\vec{a}' = \vec{b}.\vec{b}' = \vec{c}.\vec{c}' = 1$$

3] $$\vec{a}.\vec{b}' = \vec{a}.\vec{c}' = 0$$
$$\vec{b}.\vec{a}' = \vec{b}.\vec{c}' = 0$$
$$\vec{c}.\vec{a}' = \vec{c}.\vec{b}' = 0$$